

Data structure

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1. Print all the odd and even counts in the given singly linked list.

sol:

Given linked list:

77 → 88 → 22 → 11 → 66 → 27 → 55 → 65 → 18 → 10 → Null

Element odd/Even

77 odd

88 Even

22 Even

11 odd

66 Even

27 odd

55 odd

65 odd

18 Even

10 Even

count:

* odd elements = 5 (77, 11, 27, 55, 65)

* Even elements = 5 (88, 22, 66, 18, 10)

output:

odd count=5

Even count=5

2. search linearly for key element $k=20$ in the singly linked list.

Given linked List

77 $\rightarrow$ 88 $\rightarrow$ 22 $\rightarrow$ 11 $\rightarrow$ 66 $\rightarrow$ 27 $\rightarrow$ 55 $\rightarrow$ 65 $\rightarrow$ 18 $\rightarrow$ 10 $\rightarrow$ null

Key Element : $k=20$

Process:

- compare 20 with 77 $\rightarrow$ not equal
- * compare 20 with 88 $\rightarrow$ not equal
- * compare 20 with 22 $\rightarrow$ not equal
- * compare 20 with 11 $\rightarrow$ not equal
- * compare 20 with 66 $\rightarrow$ not equal
- * compare 20 with 27 $\rightarrow$ not equal
- * compare 20 with 55 $\rightarrow$ not equal
- * compare 20 with 65 $\rightarrow$ not equal
- * compare 20 with 18 $\rightarrow$ not equal
- * compare 20 with 10 $\rightarrow$ not equal

Since 20 is not found in any node, the search ends.

Output:

No